WORKED SOLUTION

STEP I Specimen paper, Question 8

Pure • Integral Equations

STEP I Specimen paper, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

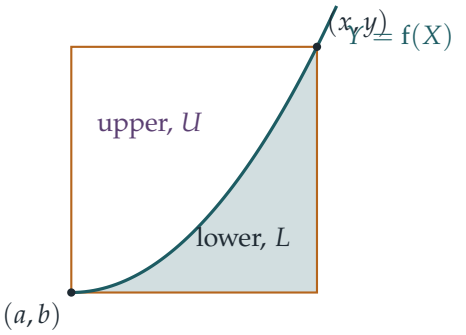
step.maths.org/questions/spec-s1-q8

The key idea

The area condition holds for *every* position of (x, y), so it is really an equation between two functions of x: one side is an integral of f and the other involves $\mathrm{f}(x)$ itself. Differentiating with respect to x removes the integral and leaves a first-order differential equation, which separates (or, more neatly, becomes the derivative of a quotient being zero).

Solution

Since $\mathrm{f}'(x) \geq 0$, the curve rises from the corner (a, b) to the opposite corner (x, y) of the rectangle, so the rectangle has width $x - a$ and height $y - b = \mathrm{f}(x) - b \geq 0$. The lower portion is the region between the curve and the bottom edge $Y = b$ (shaded below); the upper portion is the rest.



Using t as the variable of integration (since x is now the upper limit),

$$L = \int_a^x (\mathrm{f}(t) - b)\,\mathrm{d}t, \qquad L + U = (x - a)(\mathrm{f}(x) - b).$$

The condition $L = \frac{1}{2}U$ means $U = 2L$, so the whole rectangle has area $3L$:

$$3\int_a^x (\mathrm{f}(t) - b)\,\mathrm{d}t = (x-a)(\mathrm{f}(x) - b) \qquad \text{for all } x \geq a. \tag{$*$}$$

Ask yourself How do we get from an equation involving an integral to something we can solve for f?

Because $(*)$ holds for every $x \geq a$, both sides are the same function of x, so their derivatives agree. The function f is differentiable (we are told about f′), hence continuous, so by the fundamental theorem of calculus the left-hand side has derivative $3(\mathrm{f}(x) - b)$. Differentiating the right-hand side by the product rule,

$$3(\mathrm{f}(x) - b) = (\mathrm{f}(x) - b) + (x-a)\,\mathrm{f}'(x),$$

that is,

$$(x-a)\,\mathrm{f}'(x) = 2(\mathrm{f}(x) - b).$$

To tidy this, write $g(x) = \mathrm{f}(x) - b$ and $u = x - a$, so $g' = \mathrm{f}'$ and the equation is $u\,g' = 2g$. One could separate variables, $g'/g = 2/u$, but that divides by g, which might vanish. It is cleaner to notice that, for $x > a$ (so $u > 0$),

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{g}{u^2}\right) = \frac{u^2 g' - 2u\,g}{u^4} = \frac{u\,g' - 2g}{u^3} = 0.$$

So g/u^2 is a constant, k say, on $x > a$, and

$$\mathrm{f}(x) - b = k(x-a)^2 \qquad (x > a),$$

which also holds at $x = a$ since (a, b) is on the curve. As $\mathrm{f}(x) \geq b$ for $x \geq a$ (the function is non-decreasing from $\mathrm{f}(a) = b$), we have $k \geq 0$, and $k > 0$ unless the curve is the horizontal line $y = b$, for which every rectangle collapses to zero area and the condition is empty.

Thus the curve is $y - b = k(x-a)^2$ with $x \geq a$: half of the parabola with vertical axis $x = a$ and vertex at (a, b). Since $x \geq a$ we only get the right-hand arm, so it is an arc of the parabola, as required.

Conversely, this curve does satisfy the condition: $L = \int_a^x k(t-a)^2\,\mathrm{d}t = \frac{1}{3}k(x-a)^3$, while the rectangle has area $(x-a)\cdot k(x-a)^2 = k(x-a)^3$, so $U = \frac{2}{3}k(x-a)^3 = 2L$.

Checks

Take $a = b = 0$, $k = 1$ and $x = 3$: the rectangle is 3×9, area 27, and $\int_0^3 t^2\,\mathrm{d}t = 9$, so the portions are 9 and 18, ratio $1 : 2$. More generally, for $y - b = k(x-a)^p$ the lower-to-upper ratio is $1/p$ (we checked $p = \frac{1}{2}, 1, \frac{3}{2}, 2, 3$ numerically by Simpson's rule), so the ratio $\frac{1}{2}$ singles out $p = 2$; a straight line ($p = 1$) splits the rectangle equally, as it should.

Common slips

- Using x both as the limit and as the variable of integration, which makes the differentiation step confusing.
- Swapping the portions and getting $U = \frac{1}{2}L$, which leads to $(x-a)\mathrm{f}' = \frac{1}{2}(\mathrm{f}-b)$ and the curve $y - b = k\sqrt{x-a}$, a parabola with a *horizontal* axis and the wrong orientation.
- Dividing by $\mathrm{f}(x) - b$ without comment; the quotient argument above avoids this.

Beyond STEP

Equation $(*)$ is an *integral equation*: the unknown function appears inside an integral. Turning it into a differential equation by differentiating, and remembering that the original equation also fixes the initial value, is a standard first move, and the uniqueness of the answer rests on the fact that a function with zero derivative on an interval is constant (a consequence of the mean value theorem). Abbott's *Understanding Analysis* proves both the fundamental theorem of calculus and the mean value theorem carefully; Boyce and DiPrima, *Elementary Differential Equations*, covers equations like $u\,g' = 2g$.

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